

Gram-Schmidt Orthonormalization Process

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Abstract

In this talk, I will begin by informally using graphical representations for the concept of a vector and then explain how to project a given vector onto a subspace. Afterward, I will provide another graphical representation as to how to create a set of orthogonal vectors using this method of projection, given an already existing set of linearly independent vectors for some inner product space. Then I will move to an algebraic standpoint in order to introduce rigour. Once the ideas are explained and after I have defined additional information, the procedure known as the Gram-Schmidt Process will be given formally and proved. Afterwards, a general example will be demonstrated on how to computationally use the Gram-Schmidt method followed by the application of rotational matrices. While there will be mentions of geometric interpretations of concepts used and defined in this paper, the main focus is the algebra behind why the Gram-Schmidt process works, which in turn will give a deeper geometric understanding. This includes very detailed algebraic proofs and computations.

Introduction

Definition 1.1 [6]

Let V be a vector space. An inner product is a map $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ which satisfies the following properties for every $x, y, z \in V$ and $\alpha \in \mathbb{R}$:

- (i) $\langle x, x \rangle \geq 0$ for all $x \in V$ with $\langle x, x \rangle = 0$ iff $x = 0$.
- (ii) $\langle x + z, y \rangle = \langle x, y \rangle + \langle z, y \rangle$ and $\langle \alpha x, y \rangle = \alpha \langle x, y \rangle$.
- (iii) $\langle x, y \rangle = \langle y, x \rangle$.

A vector space V equipped with an inner product is called an inner product space.

Definition 1.2 [6]

Two vectors x, y in an inner product space V are called orthogonal if $\langle x, y \rangle = 0$. A subset $S \subset V$ is called an orthogonal set if all the elements in S are mutually orthogonal.

Definition 1.3 [6]

A normed space is a vector space V with a map $\|\cdot\| : V \rightarrow [0, \infty)$ which satisfies the following properties for every $x, y \in V$ and $\alpha \in \mathbb{R}$.

- (i) $\|x\| = 0$ iff $x = 0$.
- (ii) $\|\alpha x\| = |\alpha| \|x\|$.
- (iii) $\|x\| = \sqrt{\langle x, x \rangle}$.

We call $\|x\|$ the norm of x .

Definition 1.4 [2]

Let $v_1, v_2, \dots, v_n$ be vectors in a vector space V .

A sum of the form

$$\sum_{i=1}^n \alpha_i v_i$$

where $\alpha_i \in \mathbb{R}$ is called a linear combination of $v_1, \dots, v_n$.

The set of all linear combinations of $v_1, \dots, v_n$ is called the span of $v_1, \dots, v_n$ and may be denoted as

$$\text{span}(v_1, \dots, v_n).$$

Definition 1.5

The vector $u \in s := \text{span}(v_1, \dots, v_n)$ such that $x - u$ is orthogonal to s is called the projection of x onto s .

We write $\text{proj}_s x$, and it is referred to as the component of x onto s .

In the case where s is 1-dimensional, $s = \text{span}(v_1) = \{\alpha v_1 : \alpha \in \mathbb{R}\}$ and may be written as $\text{proj}_{v_1} x$.

For the case that $s = \text{span}(v_1, \dots, v_n)$ is multi-dimensional, an algebraic formula for $\text{proj}_s x$ is

$$\frac{\langle x, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1 + \dots + \frac{\langle x, v_n \rangle}{\langle v_n, v_n \rangle} v_n$$

For the single dimensional case where $s = \text{span}(v_1)$, an algebraic formula for $\text{proj}_s x$ is

$$\frac{\langle x, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1$$

Additionally, it is always the case that $\langle x - \text{proj}_{v_1} x, \text{proj}_{v_1} x \rangle = 0$.¹

¹See 1.1A in the appendix

If $\{v_1, v_2, \dots, v_n\}$ is an orthogonal set and $s = \text{span}(v_1, \dots, v_n)$, then for a vector x , $\langle x - \text{proj}_s x, \text{proj}_s x \rangle = 0$, whether or not $x \in \text{span}(v_1, \dots, v_n)$.²

However, if $\{v_1, v_2, \dots, v_n\}$ is not an orthogonal set, then for a vector x , $\langle x - \text{proj}_s x, \text{proj}_s x \rangle \neq 0$, whether or not $x \in \text{span}(v_1, \dots, v_n)$.³

The essential idea behind the Gram-Schmidt method is that given some vectors, say x_1, x_2, x_3 in $\mathbb{R}^3$ such that they are linearly independent of each other we will be recursively taking projections using these vectors until we find a set of orthogonal vectors which forms a basis for $\mathbb{R}^3$. However, generally the initial set of linearly independent vectors do not need to be a basis and the resulting set of orthogonal vectors need not form a basis either. Once we have the set of orthogonal vectors, we compute the corresponding set of orthonormal vectors. Then $\text{span}(x_1, x_2, x_3) = \text{span}(u_1, u_2, u_3)$ where $\{u_1, u_2, u_3\}$ is the set of orthonormal vectors.

Let's begin with the projection of x_2 along x_1 and call it $\text{proj}_{x_1} x_2$.

Geometrically, $\text{proj}_{x_1} x_2$ is parallel to x_1 , since it only differs by a scalar, and $x_2 - \text{proj}_{x_1} x_2$ is perpendicular to x_1 .

To simplify this we are going to be assigning some values.

Let $v_1 = x_1$ and let $v_2 = x_2 - \text{proj}_{x_1} x_2$. Notice here that v_1 and v_2 together span a plane, a subspace and so to find a vector orthogonal to these, we need to find a vector orthogonal to this subspace spanned by v_1 and v_2 .

Let s denote the subspace spanned by v_1 and v_2 . Then the projection of x_3 along s is $\text{proj}_s x_3$. It then follows that $x_3 - \text{proj}_s x_3$ is orthogonal to $\text{proj}_s x_3$ ⁴ and so orthogonal to both v_1 and v_2 .

Let $v_3 = x_3 - \text{proj}_s x_3$. Then $\{v_1, v_2, v_3\}$ is an orthogonal set of vectors which forms a basis for $\mathbb{R}^3$ and computing the unit vector for each yields an orthonormal basis for $\mathbb{R}^3$.

Gram-Schmidt Process (unmodified) 1.6 [2]

Let $\{x_1, x_2, \dots, x_n\}$ be a linearly independent set in an inner product space V .

Let $u_1 = \frac{x_1}{\|x_1\|}$ and define $u_2, \dots, u_n$ recursively by $u_{k+1} = \frac{x_{k+1} - P_k}{\|x_{k+1} - P_k\|}$

where $P_k = \langle x_{k+1}, u_1 \rangle u_1 + \dots + \langle x_{k+1}, u_k \rangle u_k$.

Then a set of orthonormal vectors in V exists and is $\{u_1, \dots, u_n\}$.

Additionally, $\text{span}(u_1, \dots, u_n) = \text{span}(x_1, \dots, x_n)$.

There is also a modified version for simpler computation where the difference from that of the unmodified procedure is that the unit vector computation is performed last instead of during the computation of an orthogonal vector. Since the resulting vectors in the modified process are still orthogonal, the procedures are equivalent. In order to verify this, the proof would be exactly the same except that the inner product of a vector with itself would not necessarily be 1 if the vector is not necessarily a unit vector.

²See 1.2A in the appendix

³See 1.3A in the appendix

⁴It is important to note this is only the case because the subspace s is spanned by orthogonal vectors, v_1 and v_2 .

Gram-Schmidt Process, computational (modified) 1.7

Suppose $\{x_1, \dots, x_n\}$ is a linearly independent set of vectors in an inner product space V .

Let $v_1 = x_1$.

Let $v_k = x_k - \sum_{i=1}^{k-1} \frac{\langle x_k, v_i \rangle}{\langle v_i, v_i \rangle} v_i$.

Let $u_k = \frac{v_k}{\|v_k\|}$.

Then $\{u_1, \dots, u_n\}$ is an orthonormal set in V .

Additionally, $\text{span}(u_1, \dots, u_n) = \text{span}(x_1, \dots, x_n)$.

Proof for Gram-Schmidt Process 1.8

This proof will be using strong induction on the proposition "there exists an orthonormal set, namely $\{u_1, u_2, \dots, u_n\}$, given a linearly independent set of vectors $\{x_1, x_2, \dots, x_n\}$ for an inner product space V " by using the construction from 1.6. After which, we will prove that $\text{span}(x_1, x_2, \dots, x_n) = \text{span}(u_1, u_2, \dots, u_n)$. We will first look at a couple of base cases.

Base Case 1:

Let $\{x_1, x_2\}$ be a set of linearly independent vectors for an inner product space V .

Let $u_1 = \frac{x_1}{\|x_1\|}$ and $u_2 = \frac{x_2 - P_1}{\|x_2 - P_1\|}$ where $P_1 = \langle x_2, u_1 \rangle u_1$.

$$\begin{aligned} \langle u_1, u_2 \rangle &= \langle u_1, \frac{x_2 - P_1}{\|x_2 - P_1\|} \rangle = \frac{\langle u_1, x_2 \rangle}{\|x_2 - P_1\|} - \frac{\langle u_1, P_1 \rangle}{\|x_2 - P_1\|} \\ &= \frac{\langle u_1, x_2 \rangle}{\|x_2 - P_1\|} - \frac{\langle u_1, \langle x_2, u_1 \rangle u_1 \rangle}{\|x_2 - P_1\|} \\ &= \frac{\langle u_1, x_2 \rangle}{\|x_2 - P_1\|} - \frac{\langle x_2, u_1 \rangle \langle u_1, u_1 \rangle}{\|x_2 - P_1\|} = 0 \text{ since } \langle u_1, u_1 \rangle = 1. \end{aligned}$$

So there exists an orthonormal set from these constructions, $\{u_1, u_2\}$ in V .

Base Case 2:

Let $\{x_1, x_2, x_3\}$ be a set of linearly independent vectors for an inner product space V .

Let $u_1 = \frac{x_1}{\|x_1\|}$, $u_2 = \frac{x_2 - P_1}{\|x_2 - P_1\|}$ and $u_3 = \frac{x_3 - P_2}{\|x_3 - P_2\|}$ where $P_1 = \langle x_2, u_1 \rangle u_1$, $P_2 = \langle x_3, u_1 \rangle u_1 + \langle x_3, u_2 \rangle u_2$.

Since $\{x_1, x_2, x_3\}$ is linearly independent, $\{x_1, x_2\}$ is as well and so it follows from the first base case

that $\langle u_1, u_2 \rangle = 0$.

$$\begin{aligned} \langle u_2, u_3 \rangle &= \langle u_2, \frac{x_3 - P_2}{\|x_3 - P_2\|} \rangle = \frac{\langle u_2, x_3 \rangle}{\|x_3 - P_2\|} - \frac{\langle u_2, P_2 \rangle}{\|x_3 - P_2\|} \\ &= \frac{\langle u_2, x_3 \rangle}{\|x_3 - P_2\|} - \frac{\langle u_2, \langle x_3, u_1 \rangle u_1 + \langle x_3, u_2 \rangle u_2 \rangle}{\|x_3 - P_2\|} \\ &= \frac{\langle u_2, x_3 \rangle}{\|x_3 - P_2\|} - \frac{(\langle x_3, u_1 \rangle \langle u_2, u_1 \rangle + \langle x_3, u_2 \rangle \langle u_2, u_2 \rangle)}{\|x_3 - P_2\|} \\ &= \frac{\langle u_2, x_3 \rangle}{\|x_3 - P_2\|} - \frac{\langle x_3, u_2 \rangle}{\|x_3 - P_2\|} = 0 \text{ since } \langle u_2, u_1 \rangle = 0 \text{ and } \langle u_2, u_2 \rangle = 1. \\ \langle u_1, u_3 \rangle &= \langle u_1, \frac{x_3 - P_2}{\|x_3 - P_2\|} \rangle = \frac{\langle u_1, x_3 \rangle}{\|x_3 - P_2\|} - \frac{\langle u_1, P_2 \rangle}{\|x_3 - P_2\|} \\ &= \frac{\langle u_1, x_3 \rangle}{\|x_3 - P_2\|} - \frac{\langle u_1, \langle x_3, u_1 \rangle u_1 + \langle x_3, u_2 \rangle u_2 \rangle}{\|x_3 - P_2\|} \\ &= \frac{\langle u_1, x_3 \rangle}{\|x_3 - P_2\|} - \frac{(\langle x_3, u_1 \rangle \langle u_1, u_1 \rangle + \langle x_3, u_2 \rangle \langle u_1, u_2 \rangle)}{\|x_3 - P_2\|} \\ &= \frac{\langle u_1, x_3 \rangle}{\|x_3 - P_2\|} - \frac{\langle x_3, u_1 \rangle}{\|x_3 - P_2\|} = 0 \text{ since } \langle u_1, u_1 \rangle = 1 \text{ and } \langle u_1, u_2 \rangle = 0. \end{aligned}$$

So there exists an orthonormal set from these constructions, $\{u_1, u_2, u_3\}$.

Inductive Hypotheses:

Let $m \in N$ and $\{x_1, x_2, x_3, \dots, x_m\}$ be a set of linearly independent vectors in an inner product space V and suppose that there exists an orthonormal set $\{u_1, u_2, \dots, u_m\}$ in V such that $u_1 = \frac{x_1}{\|x_1\|}$ and $u_k = \frac{x_k - P_{k-1}}{\|x_k - P_{k-1}\|}$ if $(k > 1) \in N$. Now suppose that $\{x_1, x_2, \dots, x_{m+1}\}$ is also a set of linearly independent vectors in V .

Let $P_k = \langle x_{k+1}, u_1 \rangle u_1 + \dots + \langle x_{k+1}, u_k \rangle u_k$ if $k \geq 1$ and let $(1 \leq i \leq m) \in N$.

$$\begin{aligned} \langle u_i, u_{m+1} \rangle &= \langle u_i, \frac{x_{m+1} - P_m}{\|x_{m+1} - P_m\|} \rangle = \frac{\langle u_i, x_{m+1} \rangle}{\|x_{m+1} - P_m\|} - \frac{\langle u_i, P_m \rangle}{\|x_{m+1} - P_m\|} \\ &= \frac{\langle u_i, x_{m+1} \rangle}{\|x_{m+1} - P_m\|} - \frac{\langle u_i, \langle x_{m+1}, u_1 \rangle u_1 + \dots + \langle x_{m+1}, u_m \rangle u_m \rangle}{\|x_{m+1} - P_m\|} \\ &= \frac{\langle u_i, x_{m+1} \rangle}{\|x_{m+1} - P_m\|} - \frac{(\langle x_{m+1}, u_1 \rangle \langle u_i, u_1 \rangle + \dots + \langle x_{m+1}, u_m \rangle \langle u_i, u_m \rangle)}{\|x_{m+1} - P_m\|} \end{aligned}$$

By the inductive hypothesis, $\{u_1, u_2, \dots, u_m\}$ is an orthonormal set and since $(1 \leq i \leq m)$, it follows that $u_i \in \{u_1, \dots, u_m\}$. In fact, $\{u_j : 1 \leq j \leq m\} = \{u_1, \dots, u_m\}$. If $(1 \leq w \leq m) \in N$, then $\langle u_i, u_w \rangle = 0$ when $i \neq w$ and $\langle u_i, u_w \rangle = 1$ when $i = w$.

So the sum

$$\langle u_i, u_1 \rangle + \langle u_i, u_2 \rangle + \dots + \langle u_i, u_m \rangle = \sum_{l=1}^m \langle u_i, u_l \rangle = 1$$

and the sum

$$\sum_{l \in \{1, 2, \dots, m\} \setminus i} \langle u_i, u_l \rangle = 0$$

because i is a unique positive integer between 1 and m , inclusively, i.e. i cannot be more than a single positive integer and has to be no less than a single positive integer between 1 and m , inclusively.

Hence

$$\frac{\langle x_{m+1}, u_1 \rangle \langle u_i, u_1 \rangle + \dots + \langle x_{m+1}, u_m \rangle \langle u_i, u_m \rangle}{\|x_{m+1} - P_m\|} = \frac{1}{\|x_{m+1} - P_m\|} \sum_{l=1}^m \langle x_{m+1}, u_l \rangle \langle u_i, u_l \rangle = \frac{\langle x_{m+1}, u_i \rangle \langle u_i, u_i \rangle}{\|x_{m+1} - P_m\|}$$

where $q = i$ and so it follows that

$$\frac{1}{\|x_{m+1} - P_m\|} \sum_{l=1}^m \langle x_{m+1}, u_l \rangle \langle u_i, u_l \rangle = \frac{\langle x_{m+1}, u_i \rangle}{\|x_{m+1} - P_m\|}$$

since $\langle u_i, u_i \rangle = 1$.

Thus

$$\frac{\langle u_i, x_{m+1} \rangle}{\|x_{m+1} - P_m\|} - \frac{(\langle x_{m+1}, u_1 \rangle \langle u_i, u_1 \rangle + \dots + \langle x_{m+1}, u_m \rangle \langle u_i, u_m \rangle)}{\|x_{m+1} - P_m\|} = \frac{\langle u_i, x_{m+1} \rangle}{\|x_{m+1} - P_m\|} - \frac{\langle x_{m+1}, u_i \rangle}{\|x_{m+1} - P_m\|} = 0$$

which holds for any $(1 \leq i \leq m)$ and so $\{u_1, u_2, \dots, u_m, u_{m+1}\}$ is an orthonormal set of vectors in V .

Therefore, given a set of linearly independent vectors, $\{x_1, x_2, \dots, x_n\}$ in an inner product space V , using the construction in 1.6, there exists a set of orthonormal vectors in V , $\{u_1, u_2, \dots, u_n\}$, by strong induction.

To show that the $\text{span}(x_1, \dots, x_n) = \text{span}(u_1, \dots, u_n)$ we need to be able to show that an arbitrary vector $y \in \text{span}(x_1, \dots, x_n)$ can be written as a linear combination in terms of $u_1, u_2, \dots, u_n$ and vice versa [8].

Let $y = a_1x_1 + \dots + a_nx_n = a_1u_1\|x_1\| + a_2\|x_2 - P_1\|u_2 + a_2P_1 + \dots + a_n\|x_n - P_{n-1}\|u_n + a_nP_{n-1} =$
 $a_1u_1\|x_1\| + a_2\|x_2 - P_1\|u_2 + a_2\langle x_2, u_1 \rangle u_1 + \dots + a_n\|x_n - P_{n-1}\|u_n + a_n\langle x_n, u_1 \rangle u_1 + \dots + a_n\langle x_n, u_{n-1} \rangle u_{n-1} =$
 $u_1(a_1\|x_1\| + a_2\langle x_2, u_1 \rangle + \dots + a_n\langle x_n, u_1 \rangle) + u_2(a_2\|x_2 - P_1\| + a_3\langle x_3, u_2 \rangle + \dots + a_n\langle x_n, u_2 \rangle) + \dots + u_{n-1}(a_{n-1}\|x_{n-1} -$
 $P_{n-2}\| + a_n\langle x_n, u_{n-1} \rangle) + u_n(a_n\|x_n - P_{n-1}\|)$ by adding like-terms of $u_1, \dots, u_n$.

Hence, y can be written from a linear combination of $x_1, \dots, x_n$ to one of $u_1, \dots, u_n$ and it is straightforward to work backwards from a linear combination of $u_1, \dots, u_n$ to one of $x_1, \dots, x_n$.

Therefore, $\text{span}(x_1, \dots, x_n) = \text{span}(u_1, \dots, u_n)$. □

Computational Example 1.9

Given three vectors in $\mathbb{R}^3$, construct an orthonormal set.

Note: This example uses 1.7 instead of 1.6.

$$x_1 = (1, 2, -2)^T, x_2 = (4, 3, 2)^T, x_3 = (1, 2, 1)^T$$

Firstly, it is important to verify that the given vectors are actually linearly independent in order for the procedure to work as intended. This will be done by reducing a matrix, where the given vectors are the columns, into the identity matrix using elementary row operations.

$$\begin{pmatrix} 1 & 4 & 1 \\ 2 & 3 & 2 \\ -2 & 2 & 1 \end{pmatrix} \xrightarrow{\substack{\text{row2} = 2 * \text{row1} + \text{row2} \\ \text{row3} = 2 * \text{row1} + \text{row3}}} \begin{pmatrix} 1 & 4 & 1 \\ 0 & -5 & 0 \\ 0 & 10 & 3 \end{pmatrix} \xrightarrow{\text{row2} = \text{row2}/5} \begin{pmatrix} 1 & 4 & 1 \\ 0 & -1 & 0 \\ 0 & 10 & 3 \end{pmatrix} \rightarrow \dots \rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\text{Let } v_1 = (1, 2, -2)^T$$

$$\text{Let } v_2 = x_2 - \frac{\langle x_2, v_1 \rangle v_1}{\langle v_1, v_1 \rangle} = (4, 3, 2)^T - \frac{2}{3}(1, 2, -2)^T = \left(\frac{10}{3}, \frac{5}{3}, \frac{10}{3}\right)^T$$

$$\begin{aligned} \text{Let } v_3 &= x_3 - \frac{\langle x_3, v_1 \rangle v_1}{\langle v_1, v_1 \rangle} - \frac{\langle x_3, v_2 \rangle v_2}{\langle v_2, v_2 \rangle} = (1, 2, 1)^T - \frac{3}{9}(1, 2, -2)^T - \frac{10}{25}\left(\frac{10}{3}, \frac{5}{3}, \frac{10}{3}\right)^T \\ &= (1, 2, 1)^T - \frac{1}{3}(1, 2, -2)^T - \frac{2}{5}\left(\frac{10}{3}, \frac{5}{3}, \frac{10}{3}\right)^T \\ &= \left(\frac{3}{3}, \frac{6}{3}, \frac{3}{3}\right)^T + \left(\frac{-1}{3}, \frac{-2}{3}, \frac{2}{3}\right)^T + \left(\frac{-4}{3}, \frac{-2}{3}, \frac{-4}{3}\right)^T \\ &= \left(\frac{3}{3}, \frac{6}{3}, \frac{3}{3}\right)^T + \left(\frac{-5}{3}, \frac{-4}{3}, \frac{-2}{3}\right)^T = \left(\frac{-2}{3}, \frac{2}{3}, \frac{1}{3}\right)^T \end{aligned}$$

Before we compute the unit vectors, it is a good idea to check that our newly constructed ones are orthogonal to each other.

$$\langle v_1, v_2 \rangle = (1, 2, -2)^T \begin{pmatrix} 10/3 \\ 5/3 \\ 10/3 \end{pmatrix} = \frac{10}{3} + \frac{10}{3} - \frac{20}{3} = 0$$

$$\langle v_2, v_3 \rangle = \left(\frac{10}{3}, \frac{5}{3}, \frac{10}{3}\right)^T \begin{pmatrix} -2/3 \\ 2/3 \\ 1/3 \end{pmatrix} = \frac{-20}{9} + \frac{10}{9} + \frac{10}{9} = 0$$

$$\langle v_1, v_3 \rangle = (1, 2, -2)^T \begin{pmatrix} -2/3 \\ 2/3 \\ 1/3 \end{pmatrix} = \frac{-2}{3} + \frac{4}{3} + \frac{-2}{3} = 0$$

This verifies that the construction done has been computed properly.

Finally, computing the unit vectors for the respective orthogonal ones, we obtain...

$$u_1 = \frac{v_1}{\|v_1\|} = \frac{v_1}{3}$$

$$u_2 = \frac{v_2}{\|v_2\|} = \frac{v_2}{5}, u_3 = \frac{v_3}{\|v_3\|} = \frac{v_3}{1}$$

This ends our example as we have now constructed an orthonormal basis for V , $\{u_1, u_2, u_3\}$, given a non-orthogonal basis for V .

Some Preliminaries

Definition 2.0 [6]

Let $A = [a_{ij}]$ be an $m \times n$ matrix. The adjoint A^* of A is the $n \times m$ matrix $B = [b_{ij}]$ such that $b_{ij} = \overline{a_{ji}}$ for all i, j .

It follows that if A is a matrix of only real numbers, then $A^* = A^T$. [6]

Theorem 2.1 [6]

Let $A = [a_{ij}]$ be an $m \times n$ matrix with real entries.

Then the adjoint matrix for A satisfies $\langle Ax, y \rangle = \langle x, A^*y \rangle$.

Definition 2.2 [6]

Let $T : H \rightarrow K$ be a linear operator. Then T is called an isometry if $\|Tx\| = \|x\|$ for every $x \in H$.

Theorem 2.3

Let $A = [a_{ij}]$ be an orthogonal matrix. Then the linear operator $T : H \rightarrow K$ such that $T(x) = Ax, x \in H$, and A is an orthogonal matrix. Then the following are true:

T is an isometry.

T also preserves the angle between vectors.

Proof

$$\|Ax\|^2 = \langle Ax, Ax \rangle = \langle x, A^*Ax \rangle = \langle x, A^T Ax \rangle = \langle x, A^{-1}Ax \rangle = \langle x, x \rangle = \|x\|^2.$$

Since $\|Ax\|^2 = \|x\|^2$, it follows that $\|Ax\| = \|x\|$ and by definition 2.2, T is an isometry.

$$\langle Ax, Ay \rangle = \langle x, A^*Ay \rangle = \langle x, A^T Ay \rangle = \langle x, A^{-1}Ay \rangle = \langle x, y \rangle.$$

$\|x\| = \|Ax\|, \|y\| = \|Ay\|$ and so $\arccos \frac{\langle x, y \rangle}{\|x\|\|y\|} = \arccos \frac{\langle Ax, Ay \rangle}{\|Ax\|\|Ay\|}$, which means the angle between x and y is the same as the angle between Ax and Ay . Thus T also preserves the angle between vectors. $\square$

An Application: Rotational Matrices

Problem:

Say you want to rotate a point, $P = (x, y, z)^T$, some degrees θ counter-clockwise about a line spanned by a vector Q .

Solution:

Do Gram-Schmidt with three vectors: Q, Q', Q'' where $\{Q, Q', Q''\}$ is a set of linearly independent vectors (Q' and Q'' are arbitrary). This creates a set of orthonormal vectors $\{u_1, u_2, u_3\}$ such that

$$u_1 = (\alpha_1, \beta_1, \gamma_1)^T, u_2 = (\alpha_2, \beta_2, \gamma_2)^T, u_3 = (\alpha_3, \beta_3, \gamma_3)^T \text{ where } u_1 \text{ is in the span of } Q.$$

$$\text{Now let } R = ABA^T, \text{ where } A = [u_1, u_2, u_3], B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos(\theta) & -\sin(\theta) \\ 0 & \sin(\theta) & \cos(\theta) \end{pmatrix}, \text{ where } B \text{ is derived by do-}$$

ing trigonometry with the unit vectors orthogonal to $(1, 0, 0)^T$. Since a product of two or more orthogonal matrices are orthogonal [5], R is orthogonal because A, A^T, B are orthogonal. So what is R and what does it do to P when we multiply them together? If we consider B then we realize that Bx is x rotated θ degrees counter-clockwise about the line spanned by $(1, 0, 0)^T$. In other words, the angle between Bx and x is exactly the angle of rotation. We can simplify the problem by treating u_1 as our new ' x -axis', however in order to do that we must change basis with $\{u_1, u_2, u_3\}$ as the new basis. So consider the equation:

$$x' u_1 + y' u_2 + z' u_3 = x e_1 + y e_2 + z e_3$$

In matrix form this is represented as

$$\begin{pmatrix} \alpha_1 & \alpha_2 & \alpha_3 \\ \beta_1 & \beta_2 & \beta_3 \\ \gamma_1 & \gamma_2 & \gamma_3 \end{pmatrix} \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

or simply $AP' = P$, where $P' = (x', y', z')^T$. Solving for P' we get that $P' = A^T P$. It then follows that the angle between BP' and P' is the angle of rotation for P' counter-clockwise about u_1 and hence about Q . So the angle between $BA^T P$ and $A^T P$ is the angle of rotation for $A^T P$ counter-clockwise about Q . However we need to rotate P counter-clockwise about Q , not $A^T P$, so we just multiply by A , which shifts everything back into the standard basis and since orthogonal matrices preserve angles between vectors, the angle of rotation does not change. Thus ABA^T rotates P a given amount of degrees counter-clockwise about Q .

Example

Let $Q = (1, -1, 9)^T$, $P = (1, 0, 0)^T$, $\theta = 37.4$.

After doing Gram-Schmidt on $\{Q, e_2, e_3\}$, we get

$$A = \begin{pmatrix} 0.10976426 & 0.01212143 & -0.99388373 \\ -0.10976426 & 0.99395765 & 0 \\ 0.98787834 & 0.10909291 & 0.11043153 \end{pmatrix}$$

$$B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0.79441462 & -0.60737584 \\ 0 & 0.60737584 & 0.79441462 \end{pmatrix}$$

$$A^T = \begin{pmatrix} 0.10976426 & -0.10976426 & 0.98787834 \\ 0.01212143 & 0.99395765 & 0.10909291 \\ -0.99388373 & 0 & 0.11043153 \end{pmatrix}$$

$$R = ABA^T = \begin{pmatrix} 0.79689155 & -0.60249037 & -0.04437577 \\ 0.5975365 & 0.79689155 & -0.08896055 \\ 0.08896055 & 0.04437577 & 0.99504614 \end{pmatrix}$$

Finally, when we multiply R and P together, we obtain

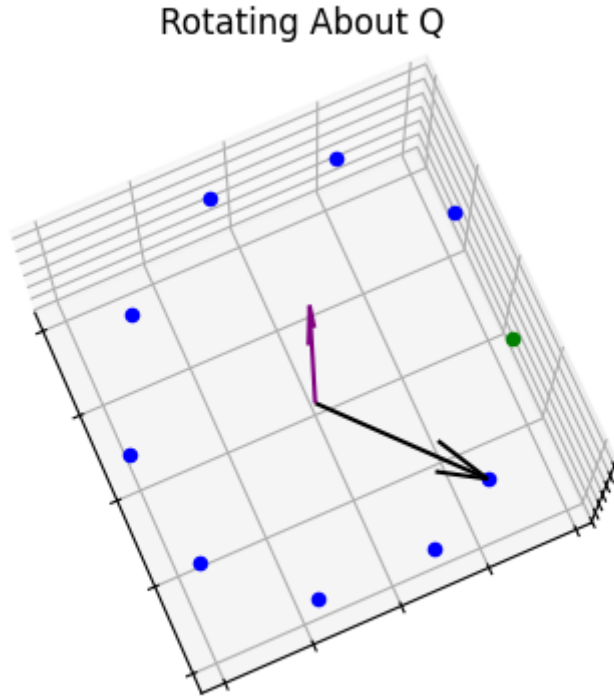
$$RP = (0.79689155, 0.5975365, 0.08896055)^T$$

To verify this is indeed the actual rotated point P , 37.4° counter-clockwise about Q , is a bit tricky because we are working in $\mathbb{R}^3$, however there actually exists a very convenient formula called Rodrigues' Rotation Formula. The inputs for the formula are k : unit vector of axis of rotation (in this case $k = \frac{Q}{\|Q\|}$), v : arbitrary vector which rotates about Q , θ : angle of rotation according to the right-hand rule (which corresponds to counter-clockwise rotation [10]), v_{rot} : the output of the formula which is the rotated vector.

$$v_{rot} = v * \cos(\theta) + (k \times v) * \sin(\theta) + k * (k \cdot v) * (1 - \cos(\theta))$$

[11].

After computing it we get $v_{rot} = (0.79689155, 0.5975365, 0.08896055)^T$, which is indeed our initial calculation. Additionally, here is a plot of P rotated about Q (the purple vector) 10 successive times. The green dot is the second rotation, so it is clear that it is rotating counter-clockwise. For the clockwise case, simply translate B before any multiplication.



Appendix

1.1A

$$\begin{aligned}\langle x - \text{proj}_{v_1} x, \text{proj}_{v_1} x \rangle &= \langle x - \frac{\langle x, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1, \frac{\langle x, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1 \rangle = \langle x, \frac{\langle x, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1 \rangle + \langle -\frac{\langle x, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1, \frac{\langle x, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1 \rangle \\ &= \frac{\langle x, v_1 \rangle}{\langle v_1, v_1 \rangle} \langle x, v_1 \rangle - \frac{\langle x, v_1 \rangle}{\langle v_1, v_1 \rangle} \frac{\langle x, v_1 \rangle}{\langle v_1, v_1 \rangle} \langle v_1, v_1 \rangle = \frac{\langle x, v_1 \rangle}{\langle v_1, v_1 \rangle} \langle x, v_1 \rangle - \frac{\langle x, v_1 \rangle}{\langle v_1, v_1 \rangle} \langle x, v_1 \rangle = 0.\end{aligned}$$

1.2A

Let $B = a_1 v_1 + \dots + a_n v_n = \sum_{i=1}^n a_i v_i$, where $a_1 = \frac{\langle x, v_1 \rangle}{\langle v_1, v_1 \rangle}, \dots, a_n = \frac{\langle x, v_n \rangle}{\langle v_n, v_n \rangle}$. Then the projection of x along a subspace s , spanned by $v_1, \dots, v_n$, $\text{proj}_s x = B$.

$$\langle x - \text{proj}_s x, \text{proj}_s x \rangle = \langle x - B, B \rangle = \langle x, B \rangle - \langle B, B \rangle.$$

$$\langle x, B \rangle = \langle x, \sum_{i=1}^n a_i v_i \rangle = a_1 \langle x, v_1 \rangle + \dots + a_n \langle x, v_n \rangle = \sum_{i=1}^n a_i \langle x, v_i \rangle = \sum_{i=1}^n \frac{\langle x, v_i \rangle}{\langle v_i, v_i \rangle} \langle x, v_i \rangle$$

$$\begin{aligned}\langle B, B \rangle &= \langle \sum_{i=1}^n a_i v_i, B \rangle = a_1 \langle v_1, B \rangle + \dots + a_n \langle v_n, B \rangle = \sum_{i=1}^n a_i \langle v_i, B \rangle = \sum_{i=1}^n a_i \sum_{j=1}^n a_j \langle v_i, v_j \rangle \\ &= \sum_{i=1}^n \sum_{j=1}^n \frac{\langle x, v_i \rangle}{\langle v_i, v_i \rangle} \frac{\langle x, v_j \rangle}{\langle v_j, v_j \rangle} \langle v_i, v_j \rangle\end{aligned}$$

$\langle x, B \rangle - \langle B, B \rangle = \sum_{i=1}^n \frac{\langle x, v_i \rangle}{\langle v_i, v_i \rangle} \langle x, v_i \rangle - \sum_{i=1}^n \sum_{j=1}^n \frac{\langle x, v_i \rangle}{\langle v_i, v_i \rangle} \frac{\langle x, v_j \rangle}{\langle v_j, v_j \rangle} \langle v_i, v_j \rangle$ and since $\{v_1, \dots, v_n\}$ is a set of orthogonal vectors $\sum_{i=1}^n \sum_{j=1}^n \frac{\langle x, v_i \rangle}{\langle v_i, v_i \rangle} \frac{\langle x, v_j \rangle}{\langle v_j, v_j \rangle} \langle v_i, v_j \rangle = \sum_{i=1}^n \frac{\langle x, v_i \rangle}{\langle v_i, v_i \rangle} \langle x, v_i \rangle$

Thus $\langle x - \text{proj}_s x, \text{proj}_s x \rangle = 0$.

Suppose $x \in \text{span}(v_1, \dots, v_n)$ and so $x = b_1 v_1 + \dots + b_n v_n$ and $\{v_1, \dots, v_n\}$ is a set of orthogonal vectors.

Then $\sum_{i=1}^n \frac{\langle x, v_i \rangle}{\langle v_i, v_i \rangle} \langle x, v_i \rangle = \sum_{i=1}^n \frac{1}{\langle v_i, v_i \rangle} (\sum_{j=1}^n b_j \langle v_j, v_i \rangle)^2$ because $\langle x, v_i \rangle = \sum_{j=1}^n b_j \langle v_j, v_i \rangle$.

$$\begin{aligned}\sum_{i=1}^n \frac{1}{\langle v_i, v_i \rangle} (\sum_{j=1}^n b_j \langle v_j, v_i \rangle)^2 &= \frac{1}{\langle v_1, v_1 \rangle} (b_1^2 \langle v_1, v_1 \rangle^2 + \dots + b_n^2 \langle v_n, v_1 \rangle^2) + \dots + \frac{1}{\langle v_n, v_n \rangle} (b_1^2 \langle v_1, v_n \rangle^2 + \dots + b_n^2 \langle v_n, v_n \rangle^2) = \\ &= \frac{1}{\langle v_1, v_1 \rangle} (b_1^2 \langle v_1, v_1 \rangle^2) + \dots + \frac{1}{\langle v_n, v_n \rangle} (b_n^2 \langle v_n, v_n \rangle^2) = \sum_{i=1}^n b_i^2 \langle v_i, v_i \rangle, \text{ likewise,}\end{aligned}$$

$$\begin{aligned}\sum_{i=1}^n \sum_{j=1}^n \frac{\langle x, v_i \rangle}{\langle v_i, v_i \rangle} \frac{\langle x, v_j \rangle}{\langle v_j, v_j \rangle} \langle v_i, v_j \rangle &= \sum_{i=1}^n \sum_{j=1}^n \frac{1}{\langle v_i, v_i \rangle} \frac{\langle v_i, v_j \rangle}{\langle v_j, v_j \rangle} \sum_{k=1}^n b_k \langle v_k, v_i \rangle \sum_{k=1}^n b_k \langle v_k, v_j \rangle = \\ &= \frac{\langle v_1, v_1 \rangle}{\langle v_1, v_1 \rangle} \frac{1}{\langle v_1, v_1 \rangle} b_1^2 \langle v_1, v_1 \rangle^2 + \dots + \frac{\langle v_n, v_n \rangle}{\langle v_n, v_n \rangle} \frac{1}{\langle v_n, v_n \rangle} b_n^2 \langle v_n, v_n \rangle^2 = \sum_{i=1}^n b_i^2 \langle v_i, v_i \rangle \text{ and so } \langle x - \text{proj}_s x, \text{proj}_s x \rangle = 0\end{aligned}$$

1.3A

Given the substitutions of 1.2A for B and $a_1, \dots, a_n$, it follows that $\langle x - \text{proj}_s x, \text{proj}_s x \rangle =$

$$\sum_{i=1}^n \frac{\langle x, v_i \rangle}{\langle v_i, v_i \rangle} \langle x, v_i \rangle - \sum_{i=1}^n \sum_{j=1}^n \frac{\langle x, v_i \rangle}{\langle v_i, v_i \rangle} \frac{\langle x, v_j \rangle}{\langle v_j, v_j \rangle} \langle v_i, v_j \rangle = - \sum_{i=1}^n \sum_{j \in \{1, \dots, n\} \setminus i} \frac{\langle x, v_i \rangle}{\langle v_i, v_i \rangle} \frac{\langle x, v_j \rangle}{\langle v_j, v_j \rangle} \langle v_i, v_j \rangle \neq 0.$$

If $x \in \text{span}(v_1, \dots, v_n)$, then $-\sum_{i=1}^n \sum_{j \in \{1, \dots, n\} \setminus i} \frac{\langle x, v_i \rangle}{\langle v_i, v_i \rangle} \frac{\langle x, v_j \rangle}{\langle v_j, v_j \rangle} \langle v_i, v_j \rangle =$

$$-\sum_{i=1}^n \sum_{j \in \{1, \dots, n\} \setminus i} \frac{\langle v_i, v_j \rangle}{\langle v_i, v_i \rangle \langle v_j, v_j \rangle} \sum_{k=1}^n b_k \langle v_k, v_i \rangle \sum_{k=1}^n b_k \langle v_k, v_j \rangle \neq 0.$$

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